Constructions With Monoidal Categories

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This chapter contains some material on constructions with monoidal categories.

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1 Moduli Categories of Monoidal Structures

1.1 The Moduli Category of Monoidal Structures on a Category

Let C be a category.

Definition 1.1.1.1. The **moduli category of monoidal structures on** C is the

category $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ defined by

$$\mathcal{M}_{\mathbb{E}_1}(\mathcal{C}) \overset{\text{def}}{=} \mathsf{pt} \times_{\mathsf{Cats}} \mathsf{MonCats}, \qquad \begin{tikzcd} \mathcal{M}_{\mathbb{E}_1}(\mathcal{C}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathsf{MonCats} \arrow[d, "\mathfrak{U}"] \\ \mathsf{pt} \arrow[r, "{[\mathcal{C}]}"'] & \mathsf{Cats}. \end{tikzcd}$$

Remark 1.1.1.2. In detail, **the moduli category of monoidal structures on** $\mathcal{C}$ is the category $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ where:

- *Objects.* The objects of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ are monoidal categories $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ whose underlying category is $\mathcal{C}$.

- *Morphisms.* A morphism from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ is a monoidal functor structure

$$\begin{aligned} \mathrm{id}^{\otimes}_{\mathcal{C}} &: A \otimes_{\mathcal{C}} B \overset{\sim}{\dashrightarrow} A \boxtimes_{\mathcal{C}} B, \\ \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}} &: \mathbb{1}_{\mathcal{C}} \overset{\sim}{\dashrightarrow} \mathbb{1}'_{\mathcal{C}} \end{aligned}$$

  on the identity functor $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ of $\mathcal{C}$.

- *Identities.* For each $M \overset{\text{def}}{=} \left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right) \in \mathrm{Obj}\left(\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})\right)$, the unit map

$$\mathbb{1}^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,M} : \mathrm{pt} \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, M)$$

  of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ at M is defined by

$$\mathrm{id}^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M} \overset{\text{def}}{=} \left(\mathrm{id}_{\otimes_{\mathcal{C}}}, \mathrm{id}_{\mathbb{1}_{\mathcal{C}}}\right),$$

  where $\left(\mathrm{id}_{\otimes_{\mathcal{C}}}, \mathrm{id}_{\mathbb{1}_{\mathcal{C}}}\right)$ is the identity monoidal functor of **??**.

- *Composition.* For each $M, N, P \in \mathrm{Obj}\left(\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})\right)$, the composition map

$$\circ^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,N,P} : \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(N, P) \times \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, N) \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}(M, P)$$

  of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ at (M, N, P) is defined by

$$\left(\mathrm{id}^{\otimes,\prime}_{\mathcal{C}}, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|\mathcal{C}}\right) \circ^{\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})}_{M,N,P} \left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right) \overset{\text{def}}{=} \left(\mathrm{id}^{\otimes,\prime}_{\mathcal{C}} \circ \mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|\mathcal{C}} \circ \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right).$$

Remark 1.1.1.3. In particular, a morphism in $\mathcal{M}_{\mathbb{E}_1}(C)$ from $\big(C, \otimes_C, \mathbb{1}_C, \alpha^C, \lambda^C, \rho^C\big)$ to $\big(C, \boxtimes_C, \mathbb{1}'_C, \alpha^{C,\prime}, \lambda^{C,\prime}, \rho^{C,\prime}\big)$ satisfies the following conditions:

1. *Naturality.* For each pair $f\colon A \to X$ and $g\colon B \to Y$ of morphisms of C, the diagram

$$\begin{tikzcd}
A \otimes_C B \arrow[r, "f \otimes_C g"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B}"'] & X \otimes_C Y \arrow[d, "\mathrm{id}^{\otimes}_{X,Y}"] \\
A \boxtimes_C B \arrow[r, "f \boxtimes_C g"'] & X \boxtimes_C Y
\end{tikzcd}$$

   commutes.

2. *Monoidality.* For each $A, B, C \in \mathrm{Obj}(C)$, the diagram

$$\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] & & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}$$

   commutes.

3. *Left Monoidal Unity.* For each $A \in \mathrm{Obj}(C)$, the diagram

$$\begin{tikzcd}
 & \mathbb{1}'_C \otimes_C A \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}'_C, A}"] & \mathbb{1}'_C \boxtimes_C A \arrow[dr, "\lambda^{C,\prime}_A"] & \\
\mathbb{1}_C \otimes_C A \arrow[ur, "\mathrm{id}^{\otimes}_{\mathbb{1}} \otimes_C \mathrm{id}_A"] \arrow[rrr, "\lambda^{C}_A"'] & & & A
\end{tikzcd}$$

   commutes.

4. *Right Monoidal Unity.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & A \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{A,\mathbb{1}'_{\mathcal{C}}}"] & A \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[dr, "\rho^{\mathcal{C},\prime}_{A}"] & \\
A \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[ur, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[rrr, "\rho^{\mathcal{C}}_{A}"'] & & & A
\end{tikzcd}$$

commutes.

Proposition 1.1.1.4. Let $\mathcal{C}$ be a category.

1. *Extra Monoidality Conditions.* Let $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ be a morphism of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$.

   (a) The diagram

$$\begin{tikzcd}
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"] \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"] \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C
\end{tikzcd}$$

   commutes.

   (b) The diagram

$$\begin{tikzcd}
A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"] \\
A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)
\end{tikzcd}$$

   commutes.

2. *Mixed Associators.* Let $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ and $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ be monoidal structures on $\mathcal{C}$ and let

$$\mathrm{id}^{\otimes}_{-_1,-_2} \colon -_1 \otimes_{\mathcal{C}} -_2 \longrightarrow -_1 \boxtimes_{\mathcal{C}} -_2$$

   be a natural transformation.

(a) If there exists a natural transformation

$$\alpha^{\otimes}_{A,B,C}\colon (A \otimes_C B) \boxtimes_C C \to A \otimes_C (B \boxtimes_C C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_C B) \otimes_C C \arrow[r, "\alpha^{C}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A\otimes_C B,C}"'] & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_C B) \boxtimes_C C \arrow[r, "\alpha^{\otimes}_{A,B,C}"'] & A \otimes_C (B \boxtimes_C C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \otimes_C B) \boxtimes_C C \arrow[r, "\alpha^{\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_C \mathrm{id}_C"'] & A \otimes_C (B \boxtimes_C C) \arrow[d, "\mathrm{id}_{A,B\boxtimes_C C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[r, "\alpha^{C,\prime}_{A,B,C}"'] & A \boxtimes_C (B \boxtimes_C C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

(b) If there exists a natural transformation

$$\alpha^{\boxtimes}_{A,B,C}\colon (A \boxtimes_C B) \otimes_C C \to A \boxtimes_C (B \otimes_C C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_C B) \otimes_C C \arrow[r, "\alpha^{C}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}^{\otimes}_{A,B\otimes_C C}"] \\
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"'] & A \boxtimes_C (B \otimes_C C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A\boxtimes_C B,C}"'] & A \boxtimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \boxtimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[r, "\alpha^{C,\prime}_{A,B,C}"'] & A \boxtimes_C (B \boxtimes_C C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

(c) If there exists a natural transformation

$$\alpha^{\boxtimes,\otimes}_{A,B,C} \colon (A \boxtimes_C B) \otimes_C C \to A \otimes_C (B \boxtimes_C C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_C B) \otimes_C C \arrow[r, "\alpha^{C}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"'] & A \otimes_C (B \boxtimes_C C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] & A \otimes_C (B \boxtimes_C C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[r, "\alpha^{C,\prime}_{A,B,C}"'] & A \boxtimes_C (B \boxtimes_C C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

Proof. *Item 1, Extra Monoidality Conditions*: We claim that Items 1a and 1b are indeed true:

1. *Proof of Item 1a:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms $\mathrm{id}^{\otimes}_{A,B}$ and id_C.

2. *Proof of Item 1b:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms id_A and $\mathrm{id}^{\otimes}_{B,C}$.

This finishes the proof.
Item 2, Mixed Associators: We claim that Items 2a to 2c are indeed true:

1. *Proof of Item 2a:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of

Item 2 of Remark 1.1.1.3 as follows:

$$
\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_C B, C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[ddd, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] & (1) & A \otimes_C (B \otimes_C C) \arrow[ddd, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
 & (A \otimes_C B) \boxtimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{\otimes}_{A,B,C}"] & (2) \\
 & (3) & \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}
$$

Since:

- Subdiagram (1) commutes by Item 1a of Item 1.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by assumption.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

2. *Proof of Item 2b:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of

Item 2 of Remark 1.1.1.3 as follows:

$$
\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[dd, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] \arrow[dr, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[rr, phantom, "(1)"] & & A \otimes_C (B \otimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_C C}"] \arrow[dd, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
 & A \boxtimes_C (B \otimes_C C) \arrow[dd, "\mathrm{id}_A \boxtimes_C \mathrm{id}^{\otimes}_{B,C}"] & \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] \arrow[rr, phantom, "(2)"] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \arrow[rr, phantom, "(3)"] & \\
 & A \boxtimes_C (B \boxtimes_C C) & &
\end{tikzcd}
$$

Since:

- Subdiagram (1) commutes by assumption.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by Item 1b of Item 1.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

3. *Proof of Item 2c:* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of

Item 2 of Remark 1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[dd, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] \arrow[ddrr, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[rr, phantom, "(1)"] & & A \otimes_C (B \otimes_C C) \arrow[dd, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
 & & \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] \arrow[rr, phantom, "(2)"] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}$$

Since subdiagrams (1) and (2) commute by assumption, it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

This finishes the proof. □

1.2 The Moduli Category of Braided Monoidal Structures on a Category

1.3 The Moduli Category of Symmetric Monoidal Structures on a Category

2 Moduli Categories of Closed Monoidal Structures

3 Moduli Categories of Refinements of Monoidal Structures

3.1 The Moduli Category of Braided Refinements of a Monoidal Structure

Appendices

A Other Chapters

Sets

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Relations

6. Relations
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Category Theory

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Bicategories

11. Types of Morphisms in Bicategories